FINAL PROJECT

FROM DATA TO PREDICTION

A complete supervised Machine Learning project

Problem formulation • Data analysis • Model development • Evaluation

FINANCE	MEDICINE	VETERINARY
Market direction	Tumour diagnosis	Cattle disease

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Project instructions, technical requirements and assessment framework

1. Project overview

The objective is to complete a small but rigorous supervised Machine Learning study on a real dataset. You will choose one of the three tracks below, formulate a prediction problem, build a reproducible workflow, compare several algorithms and defend your methodological decisions.

The central expectation

Do not treat this project as a competition to obtain the highest score. The quality of the reasoning, the validity of the experimental protocol and the interpretation of the results matter more than a single performance number.

Learning outcomes

- Formulate a real-world question as a supervised classification task, with a clearly defined prediction target and prediction time.
- Inspect a dataset, describe its variables, identify data-quality issues and use exploratory analysis to motivate decisions.
- Build a leakage-safe preprocessing and modelling pipeline appropriate to numerical and categorical features.
- Implement and compare Logistic Regression, Decision Tree, Random Forest and K-Nearest Neighbors (KNN), using a simple baseline for reference.
- Select hyperparameters using a validation strategy that respects the structure of the data.
- Evaluate the final model with appropriate metrics, inspect errors and communicate limitations.

Project format

Item	Requirement
Participants	Individual project, unless the instructor explicitly assigns teams.
Track	Choose exactly one: Finance, Medicine/Biology or Veterinary health.
Main task	Binary supervised classification.
Algorithms	Dummy baseline + Logistic Regression + Decision Tree + Random Forest + KNN.
Presentation	15 minutes total: 12 minutes to present and 3 minutes for questions.
Submission	One completed notebook (.ipynb) and a short slide deck (PDF or PPTX).

2. Choose one project track

The three tracks share the same learning objectives and assessment criteria. They differ in their data structure and in the methodological issues you must address.

Track	Dataset	Prediction question	Main methodological challenge
A — Finance	Daily OHLCV data for 20 S&P 500 equities and VIX	Can past market information help classify the direction of the next trading-day return?	Temporal ordering and prevention of look-ahead leakage.
B — Medicine / Biology	Breast Cancer Wisconsin (Diagnostic)	Can numerical measurements of cell nuclei distinguish malignant from benign tumours?	Class-specific errors; scaling and correlated features.
C — Veterinary health	ILRI Transboundary Cattle Diseases	Can environmental and geographic information help classify disease presence at a sampled location?	Missingness, class distribution and spatial/temporal dependence.

TRACK A — Finance: next-session market direction

Dataset: Daily OHLCV Data for 20 S&P 500 Equities and the CBOE Volatility Index (VIX), published on Zenodo. It contains daily Open, High, Low, Close, Adjusted Close and Volume records for 20 equities and the VIX. The published period is January 1982 to March 2025; individual equity histories begin at different dates.

Official source: Zenodo record 20192822

Prediction setting

Assume the prediction is made after the market close on day t. Use only information available up to and including day t to predict whether the Adjusted Close return from t to t+1 is positive. This timing must be stated in the notebook.

Target definition

For a selected ticker, sort observations chronologically and define the next-session return as r(t+1) = AdjClose(t+1) / AdjClose(t) - 1. The binary target is y(t) = 1 if r(t+1) > 0, and y(t) = 0 otherwise. Remove the final row because its next-session target is not observable.

Feature engineering to investigate

- Lagged returns (for example, 1-, 2-, 5- and 10-session returns).
- Rolling volatility computed only from past/current returns (for example, 5- and 20-session windows).
- Daily range or candle-related features derived from Open, High and Low at day t.
- Volume change or rolling volume statistics.
- VIX-derived information aligned by date, if available in the archive.

You may use another ticker from the archive, but you must state the ticker, its usable date range and the reason for selecting it. Inspect the archive manifest and filenames rather than assuming that every ticker has the same coverage.

Mandatory validation rule

Do not use a random train/test split. Keep the final period as a chronological test set. Tune models only on the earlier development period using TimeSeriesSplit or another justified walk-forward strategy. Never shuffle dates before splitting. All rolling features must be causal: they may not use future observations.

Questions to discuss

- What exactly is known at the prediction time?
- Does a model beat the DummyClassifier baseline consistently across time folds?
- Does class balance change across time?
- Why does a high classification score not establish that a trading strategy is profitable?

TRACK B — Medicine / Biology: breast tumour classification

Dataset: Breast Cancer Wisconsin (Diagnostic), available from the UCI Machine Learning Repository and through scikit-learn. It contains 569 instances, 30 continuous features and a binary diagnosis target. Features describe characteristics of cell nuclei measured from digitized fine-needle aspirate (FNA) images.

Official source: UCI Breast Cancer Wisconsin (Diagnostic)

Prediction question

Can cell-nucleus measurements be used to classify a tumour as malignant or benign? This is a pedagogical classification exercise. The model is not a clinical diagnostic system and must not be presented as one.

Target and feature rules

- Target: Diagnosis (malignant or benign). If using sklearn.load_breast_cancer, verify the target-name mapping and recode so that malignant is explicitly treated as the positive class (1).
- Exclude the sample ID from the predictors: it is an identifier, not a biological measurement.
- All 30 measurements are numerical. Examine scale differences and correlations before modelling.
- Use a stratified train/test split so the class proportions are approximately preserved.

Clinical interpretation of errors

In this exercise, a false negative means that a malignant case is classified as benign. A false positive means that a benign case is classified as malignant. These error types do not have the same interpretation. Choose and justify a primary metric; do not rely on accuracy alone. You should report recall for the malignant class explicitly.

Questions to discuss

- Why is scaling important for KNN and usually unnecessary for tree-based models?
- Are some measurements strongly correlated? What does that imply for interpretation?
- Which model has the highest malignant-class recall, and what is its precision?
- How would you explain the model's limitations to a non-technical audience?

TRACK C — Veterinary health: transboundary cattle diseases

Dataset: Transboundary Cattle Diseases, published by the International Livestock Research Institute (ILRI) in MELDATA. The dataset catalogue reports 3,533 data points collected in Ethiopia during 2012–2019. The archive contains three disease-specific workbooks: cbpp.xlsx, fmd.xlsx and lsd.xlsx.

Official source: ILRI / MELDATA dataset

The response in each disease file is binary (presence = 1, absence = 0). The catalogue describes Year_region, Altitude, Temperature, Precipitation, Relative_humidity, Wind_speed, longitude/latitude coordinates (x and y), and Weight. Consult the "variable" sheet in the workbook for the definitions and units.

Important dataset check before modelling

The published catalogue gives the total number of data points but does not provide the per-file class counts in its description. You must load the workbook, identify the response column, and report the exact number and proportion of positive and negative observations before choosing the target file. Do not assume that CBPP, FMD and LSD have the same sample size or class balance.

Target and feature rules

- Choose one disease workbook (CBPP, FMD or LSD) after inspecting its response distribution. State the choice and justify that both classes are sufficiently represented for the planned evaluation.
- Use the disease-presence response as y. Confirm the exact column name from the workbook; do not infer it from an assumed naming convention.
- Treat Year_region as categorical unless you justify another representation. Inspect its categories and their sample counts.
- Do not automatically use Weight as an ordinary predictor. The catalogue describes it as an observation weighting factor. Exclude it from X by default; explain its meaning and, as an extension, investigate a justified sample_weight approach if appropriate.
- Inspect duplicate locations, repeated region-year combinations and the distribution across years/regions. Explain how these structures could affect a random split.
- Coordinates may be informative, but can also let a model exploit spatial proximity. State whether x and y are used and discuss the implications.

Validation guidance

For the core exercise, use a stratified split and stratified cross-validation after checking the class distribution. In the discussion, explain why nearby locations or observations from the same region-year may not be independent. For an extension, perform a group-based or region/year holdout if the data structure supports it, and compare it with the random-split result. Do not claim that random cross-validation proves geographic generalization.

Questions to discuss

- Which disease file did you choose, and what were its positive and negative counts?
- Are the environmental variables informative in isolation, and what interactions might matter?
- What happens to performance when geographic coordinates are excluded?
- Would the model generalize to a new region or a future year? What evidence do you have?

3. Required workflow

Complete all steps in the order below. The notebook must contain both code and written interpretation. Screenshots of code or outputs are not a substitute for explanations.

01 — Problem formulation

Describe the application, the unit of observation, the prediction time, the target variable, the meaning of each class and the intended use of the prediction. State X and y explicitly.

02 — Data acquisition and audit

Identify the source and licence, load the raw data, report dimensions, data types, duplicate rows, missing values and target distribution. Keep a record of any filtering.

03 — Exploratory Data Analysis (EDA)

Use suitable summaries and plots. Examine class balance, feature distributions, possible outliers, relationships with the target and correlations. State at least three findings that influence later modelling decisions.

04 — Preprocessing

Handle missing values, encode categorical variables and scale numerical features when needed. Explain every transformation. Fit transformations only on training folds by placing them inside a Pipeline/ColumnTransformer.

05 — Data splitting

Create a final test set before model selection. Use stratified splitting for the medical and veterinary tracks. Use a chronological split for finance. Do not inspect the test set to choose features or hyperparameters.

06 — Baseline and model development

Train a DummyClassifier baseline, then Logistic Regression, Decision Tree, Random Forest and KNN. Use consistent train/test observations and model-appropriate preprocessing.

07 — Model selection and tuning

Define a small, meaningful hyperparameter search space. Use cross-validation on the development data only. Report the search strategy, folds, scoring metric and best parameters.

08 — Final evaluation and interpretation

Refit the selected configuration on the full development set, evaluate once on the untouched test set, compare with the baseline, inspect the confusion matrix and explain the remaining errors and limitations.

4. Experimental requirements

Models to implement

Model	Minimum requirement	Examples of parameters to investigate
DummyClassifier	Use as a reference, not as a candidate winner.	strategy: most_frequent or prior

Model	Minimum requirement	Examples of parameters to investigate
Logistic Regression	Include appropriate preprocessing and convergence settings.	C; class_weight (where justified); solver
Decision Tree	Compare a shallow and a deeper tree; visualize or inspect the selected tree.	criterion; max_depth; min_samples_split; min_samples_leaf
Random Forest	Compare forest sizes and tree complexity.	n_estimators; max_features; max_depth; min_samples_leaf
KNN	Compare neighbourhood sizes and distance weighting; scale numeric inputs.	n_neighbors; weights; metric or p

Pipelines and leakage control

- Use Pipeline and, where relevant, ColumnTransformer.
- For KNN and Logistic Regression, scale numerical variables inside the pipeline.
- For Decision Tree and Random Forest, scaling is generally unnecessary; imputation and categorical encoding may still be necessary.
- Fit imputers, scalers and encoders inside each cross-validation fold. Never fit them on the complete dataset before splitting.
- Do not remove observations solely because they are difficult to classify. Document any exclusion and justify it.

Hyperparameter selection

A compact grid is sufficient. For example, investigate tree depth and minimum leaf size for Decision Tree; number of trees, maximum features and tree depth for Random Forest; and k, distance weighting and the distance metric for KNN. You are not expected to search hundreds of combinations. Explain why your candidate values are reasonable.

5. Evaluation: what must be reported

Report at least the following on the final test set: confusion matrix, accuracy, precision, recall, F1-score and a classification report with per-class scores. Add ROC-AUC and average precision / PR-AUC when predicted probabilities or decision scores are available and both classes are represented. For imbalanced data, discuss macro-averaged metrics and balanced accuracy.

Metric	Definition (binary classification)	Interpretation
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall fraction of correct predictions; can hide poor minority-class performance.
Precision	TP / (TP + FP)	Among predicted positives, the fraction that are truly positive.
Recall / Sensitivity	TP / (TP + FN)	Among actual positives, the fraction detected by the model.
Specificity	TN / (TN + FP)	Among actual negatives, the fraction correctly rejected.

Metric	Definition (binary classification)	Interpretation
F1-score	2 × Precision × Recall / (Precision + Recall)	Harmonic mean of precision and recall; does not include TN.
Balanced accuracy	(Sensitivity + Specificity) / 2	Average of class-wise recall for binary classification.
ROC-AUC	Area under TPR versus FPR across thresholds	Measures ranking discrimination across thresholds; not a fixed-threshold score.
PR-AUC / Average Precision	Area/summary of precision-recall behaviour	Often informative when the positive class is rare; state the convention used.

Metric selection must follow the problem

Finance: consider balanced accuracy, F1 and both class recalls; do not interpret classification scores as trading profitability. Medicine: explicitly prioritize and discuss malignant-class recall alongside precision. Veterinary: select the positive disease class and justify the trade-off between missed cases and false alerts.

6. Deliverables and oral defence

A. Completed notebook

- Submit one .ipynb file with a clear filename: Lastname_Firstname_Track.ipynb.
- The notebook must run from top to bottom after the data path is configured (or clearly document any manual data-acquisition step).
- Use Markdown headings, concise explanations, labelled plots and readable code. Remove unused code and unexplained output.
- Include the exact source URL, dataset citation, selected target, feature list, split dates/groups where relevant, random seed and software versions.
- Do not include credentials, private data or unlicensed third-party materials.

B. Oral presentation — 15 minutes

Time	Content
0:00–1:30	Problem, motivation and prediction target.
1:30–3:30	Dataset, unit of observation and key EDA findings.
3:30–5:30	Preprocessing and data-splitting strategy; explain leakage prevention.
5:30–8:30	Models, pipelines and hyperparameter selection.
8:30–10:30	Final results: baseline comparison, confusion matrix and key metrics.
10:30–12:00	Interpretation, limitations and conclusion.
12:00–15:00	Questions and technical discussion.

Slides should support the explanation, not reproduce the notebook. Show a small number of meaningful figures and one comparison table. Each result shown must be explainable.

Questions you should be able to answer

- What is one observation in your dataset? What is one feature? What exactly is the label?
- Why did you choose this split strategy? What would go wrong with a random split for your track?
- Why does KNN need scaling while a tree-based model generally does not?
- What does your confusion matrix reveal that accuracy alone does not?
- How did you prevent data leakage during preprocessing and tuning?
- Why did you select this model and metric? What are the limitations of your conclusion?

7. Assessment rubric

Criterion	Evidence expected	Points
Problem formulation and dataset understanding	Clear target, unit of observation, prediction time and source; accurate description of features.	10
Exploratory analysis	Useful plots/statistics; class balance, data quality and findings interpreted.	10
Preprocessing and leakage prevention	Correct treatment of missing/categorical/scaled data; valid pipeline and split.	15
Model implementation	Baseline and four required algorithms implemented correctly and consistently.	20
Validation and hyperparameter selection	Appropriate CV strategy; justified search space and reported best parameters.	15
Evaluation and interpretation	Appropriate metrics, confusion matrix, error analysis, comparison and limitations.	15
Notebook quality and reproducibility	Clear structure, executable cells, readable code, sources and fixed random seeds where applicable.	5
Oral presentation and technical defence	Clear 12-minute explanation and ability to answer 3 minutes of technical questions.	10
TOTAL		100

Grading principle

A high score is not, by itself, evidence of a strong project. A methodologically invalid result (for example, future information leaking into a financial prediction, or preprocessing performed before splitting) will be penalized even if the reported accuracy is high.

8. Reproducibility, academic integrity and dataset citations

- Use a fixed random_state where randomness is involved, and report it.
- Do not copy another student's notebook or results. You may consult documentation, but acknowledge external code or material that materially influenced your work.
- Do not fabricate results or omit unsuccessful experiments that are important to interpretation.
- Use the dataset only under its stated licence and cite the original source.
- A conclusion must be supported by the experiment actually performed. Distinguish measured results from hypotheses and limitations.

Dataset references

Finance: Al Ridhawi, M., Haj Ali, M., & Al Osman, H. (2026). Daily OHLCV Data for 20 S&P 500 Equities and the CBOE Volatility Index (1982–2025). Zenodo. https://doi.org/10.5281/zenodo.20192822

Medicine: Wolberg, W. H., Street, W. N., & Mangasarian, O. L. Breast Cancer Wisconsin (Diagnostic). UCI Machine Learning Repository. https://doi.org/10.24432/C5DW2B

Veterinary: Gizaw, S., Dione, M., Wieland, B., & Desta, H. (2026). Transboundary cattle diseases. MELDATA, Version 1. https://hdl.handle.net/20.500.11766.1/FK2/8YFO3D

The finance and ILRI dataset catalogues specify CC BY 4.0. Check the current UCI record for the exact usage terms before redistributing the medical dataset. Dataset records and file contents can change; cite the version actually used.

Minimum standard for a complete submission

A complete submission allows another person to understand the question, reproduce the train/test protocol, run the analysis from the raw or clearly documented input data, and verify the reported metrics. If a step cannot be reproduced, explain precisely why and provide the evidence available. Keep all conclusions within the scope of the dataset and the experiment.